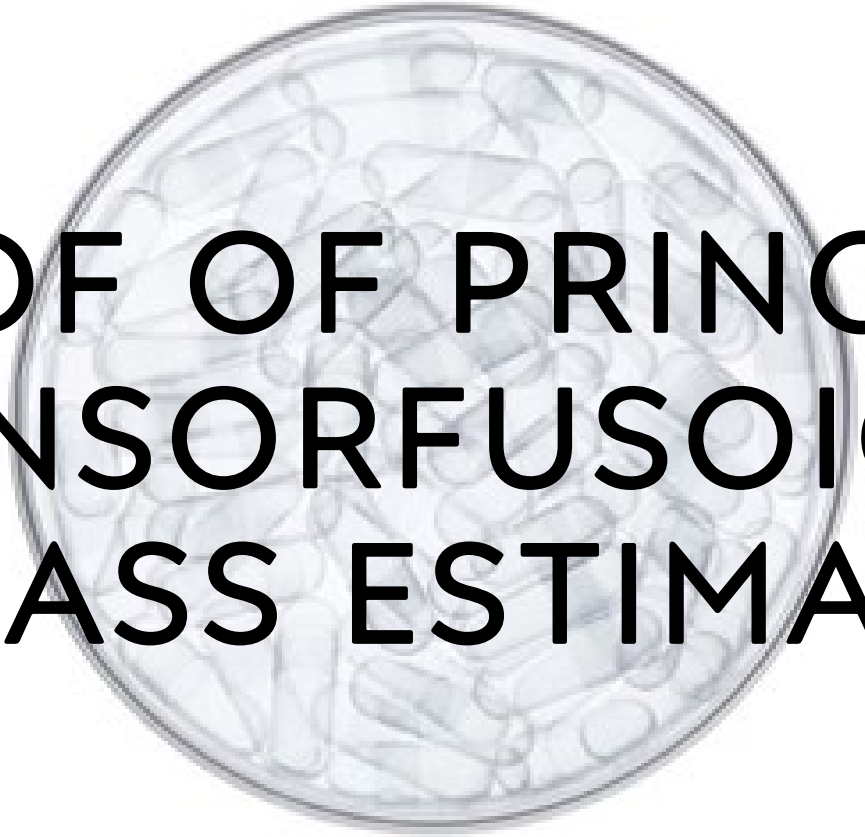


PROOF OF PRINCIPLE: SENSORFUSOION BIOMASS ESTIMATION

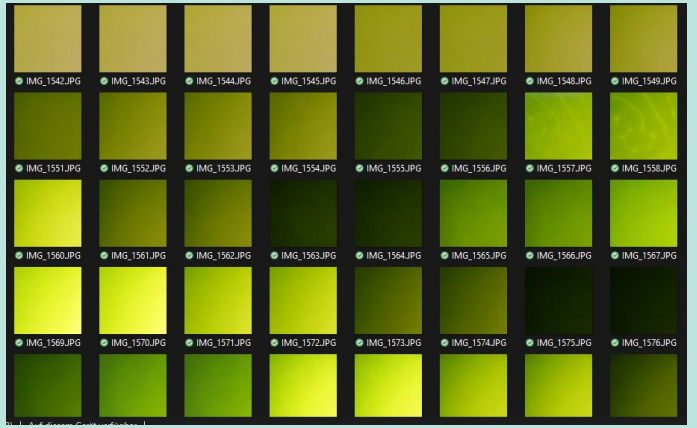
PROOF OF PRINCIPLE: SENSOR FUSION BIOMASS ESTIMATION

- Proof of Principle:
  - Estimation of biomass concentration based on optical data
- First test series:
  - Image capturing of petri dish filled with algae biomass
  - 159 pictures
  - 13 different biomass concentrations
- Goal:
  - Proof predictability of biomass concentration solely on non-invasive image data

TRAINING METHODOLOGY

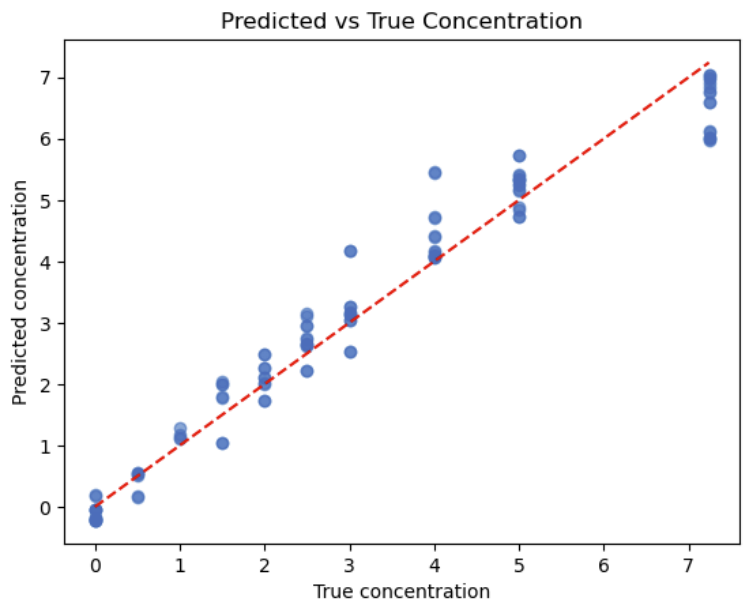
- Model setup
  - Meta regressor model
  - Combination of CNN for image processing paired together with meta neural network for meta data processing
  - Final predictions are done with combined output regressor layers
- Training process
  - 500 epochs
  - ~5min duration on GPU

- Target label
  - Biomass concentration [g/l]
- Input data
  - Meta data (e.g. probe volume, camera settings)
  - Image data:

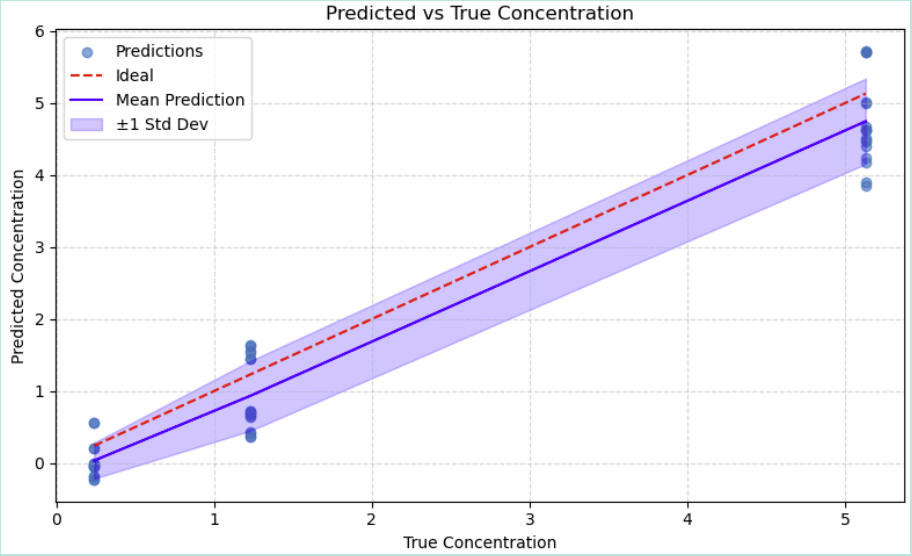


RESULTS

- Training accuracy
  - MAE* = 0.357 [g/l]



- Test accuracy on unseen data
  - MAE* = 0.504 [g/l]



*MAE = mean absolute error

FINAL COMMENTS

- Predictability of biomass concentration based on image data can be shown
- Restrictions
  - No model optimizations were done for accuracy
  - Generalization can not be judged based on single test series
- Further data collection and extension of training necessary